IMY 220

Practical 4

React Router

Due: Friday, 21 August 2026 at 08:20

The submission instructions are available on ClickUP. Any deviation from these instructions will result in a 10% deduction from your mark.

Instructions

You will create a small PhotoShare application that allows users to navigate between different pages and view individual posts.

A starter project has been supplied. The project already contains the Vite configuration, React Router dependency, supplied CSS and required folder structure. Extract the project, install its dependencies and complete the practical by modifying the supplied files.

Complete the practical using functional components, not class components.

You must use React Router to implement navigation between the different pages.

The supplied CSS must be used. You may customise the colours, but the overall layout should remain similar.

Ensure that your student number is included as a comment at the top of the `App.jsx` file. Replace uXXXXXXX with your student number.

// Student Number: uXXXXXXX

Section 1

Extract the supplied starter project.

Open the project folder in Visual Studio Code.

Install the required dependencies:

```
npm install
```

Run the development server:

```
npm run dev
```

The supplied project contains the following structure:

```
src/
├── components/
│   ├── Navigation.jsx
│   ├── Home.jsx
│   ├── Posts.jsx
│   ├── Post.jsx
│   ├── About.jsx
│   ├── Profile.jsx
│   └── NotFound.jsx
├── App.jsx
├── main.jsx
└── styles.css
```

Section 2

Create the following components:

Home
Posts
About
Profile

Each component must contain an <h1> element and at least one <p> element.

Each component must display the following content:

Home

Heading:

PhotoShare

Paragraph:

Welcome to the PhotoShare application.

Posts

Heading:

Posts

Paragraph:

View the latest PhotoShare posts.

About

Heading:

About PhotoShare

Paragraph:

PhotoShare allows users to share moments with their friends.

Profile

Heading:

My Profile

Paragraph:

View your PhotoShare profile.

Section 3

Configure React Router inside the App component.

Create the following initial routes:

URL	Component
/	Home
/posts	Posts
/about	About
/profile	Profile

Use the following React Router components:

BrowserRouter
Routes

Route

Only the component matching the current URL should be displayed.

Section 4

Create a component called `Navigation`.

The navigation must contain links to the following pages:

Home
Posts
About
Profile

Use the React Router `Link` component.

Do not use regular `<a>` elements for navigation.

The navigation must appear on every page.

Example navigation:

Home | Posts | About | Profile

Section 5

Create a component called `NotFound`.

The component must display:

404 - Page Not Found

It must also display the following paragraph:

The page you requested does not exist.

Add a catch-all route that displays the `NotFound` component when no other route matches.

Use * as the path for the catch-all route.

For example, visiting:

/random

should display the `NotFound` component.

Section 6

Inside the `Posts` component, create the following array:

```
const posts = [
  {
    id: 1,
    username: "@sarah",
    title: "Studying React today!"
  },
  {
    id: 2,
    username: "@john",
    title: "Finished Practical 4."
  },
  {
    id: 3,
    username: "@leanne",
    title: "Sunset over campus."
  }
];
```

Use `map()` to display all the posts.

Each post must display the username and the title. The title must be displayed inside a React Router `Link` component that navigates to the individual post.

Example:

@sarah
Studying React today!

Clicking the post title must navigate to:

/posts/1

The second post must navigate to:

/posts/2

The third post must navigate to:

/posts/3

Each element created using `map()` must include an appropriate `key`.

Section 7

Create a component called `Post`.

Create the following dynamic route:

/posts/:id

Use the React Router `useParams()` hook to retrieve the post ID from the URL.

The `Post` component must display:

Post Details

It must also display the selected post ID.

For example, visiting:

/posts/2

should display:

Post Details

Post ID: 2

The same `Post` component must be used for all post IDs.

Example Layout

Home page

Home | Posts | About | Profile

PhotoShare

Welcome to the PhotoShare application.

Posts page

Home | Posts | About | Profile

Posts

View the latest PhotoShare posts.

@sarah
Studying React today!

@john
Finished Practical 4.

@leanne
Sunset over campus.

Individual post page

Home | Posts | About | Profile

Post Details

Post ID: 2

Invalid route

Home | Posts | About | Profile

404 - Page Not Found

The page you requested does not exist.

Submission Files

Submit the complete Vite project.

Do not include the following folder:

node_modules

Your ZIP file should contain the complete starter project, including:

src
index.html
package.json
package-lock.json
vite.config.js

Submission Instructions

Delete the `node_modules` folder before creating your ZIP file.

ZIP the complete project folder using a tool such as 7-Zip or WinRAR.

Name the ZIP file:

p4-{student_number}.zip

Replace `{student_number}` with your student number.

Submit your ZIP file before 08:20 to avoid a late submission.